Depropanizer Column Ver 1 · OTS

DC3-V1-PI-ISO-001

Piping isometrics

Rev 0 · Issued for teaching

Unit	Depropanizer Column Ver 1 · OTS
Document number	DC3-V1-PI-ISO-001
Title	Piping isometrics
Discipline	Piping
Originator / DRAWN	Apinan Soottitantawat
Date	25-OCT-26
Source	layout model (layout.json), line list DC3-V1-PE-LIN-001, EPC P&ID rev D

Revision record

REV	DATE	DESCRIPTION	BY	CHK	APP
0	25-OCT-26	Issued for teaching	A. Soottitantawat		

Note. *Numbers come from the simulator model (design case) and the assumptions listed; not for construction. Codes are named for the method only; no client, contractor, licensor or vendor is implied.*

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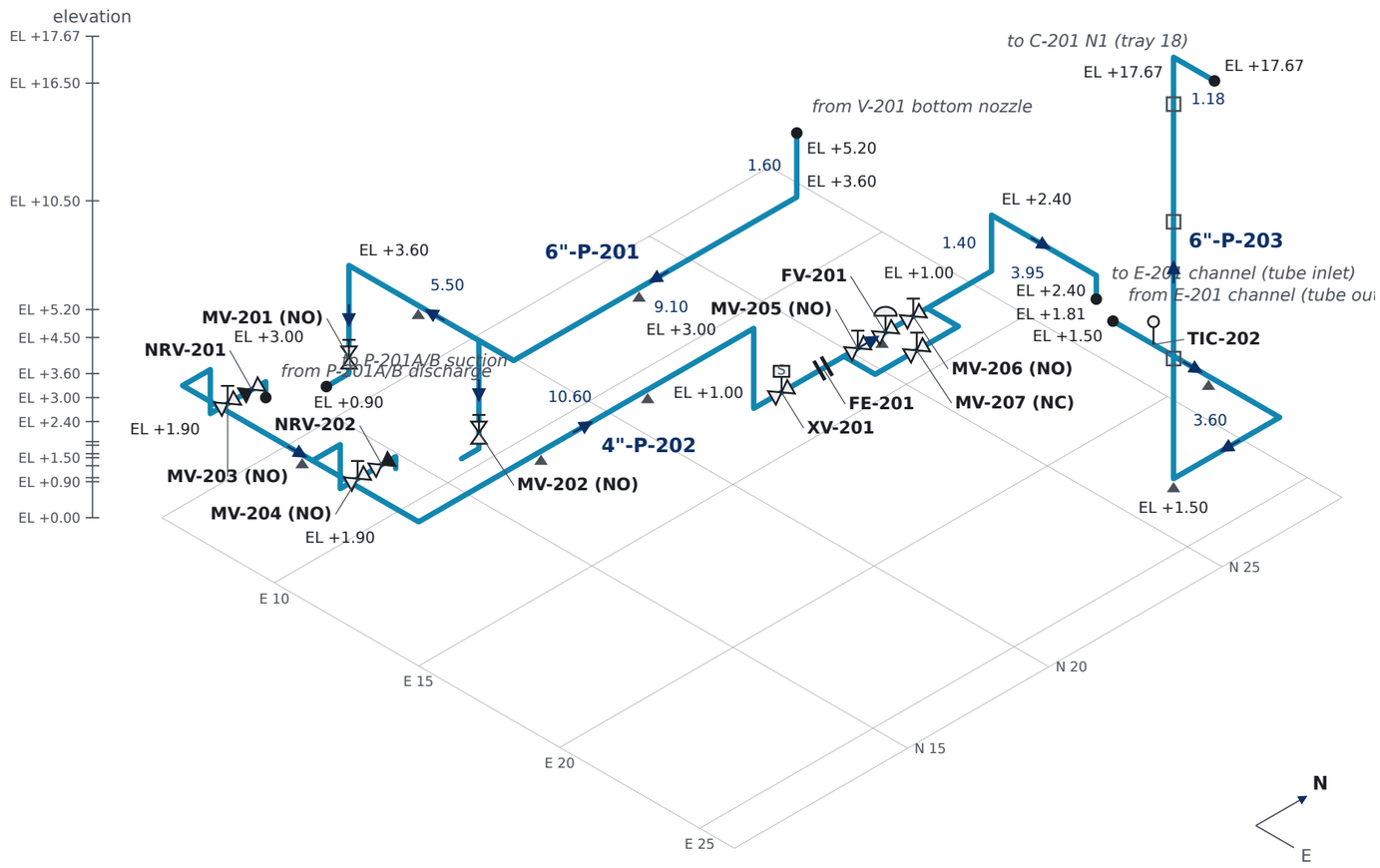
1 · Index and general notes

Sheet	Title	Lines	NPS	Class / spec	Design P / T	Route length m	On the P&ID
1	Feed: V-201 → P-201A/B → FV-201 station → E-201 → C-201 tray 18	6"-P-201, 4"-P-202, 6"-P-203	6″, 4″, 6″	150 / A1L, 300 / B1L, 300 / B1L	1.35 / 65, 2.85 / 65, 2.85 / 175	22.8, 46.2, 26.7	yes, yes, yes
2	Overhead vapour: C-201 top → E-202 (PSV-201 branch)	8"-P-204	8″	300 / B1L	1.90 / 75	34.6	yes
3	Reflux: P-202A/B → FV-204 → C-201 tray 27	4"-P-208, 4"-P-209	4″, 4″	300 / B1L, 300 / B1L	2.60 / 70, 2.60 / 70	26.8, 45.1	yes, yes
4	Bottoms: C-201 sump → LV-202 → rack → debutanizer	4"-P-210	4″	300 / B1L	1.95 / 140	61.0	yes
5	Relief: PSV-201 outlet → flare header	6"-FL-601	6″	150 / A1L	0.50 / 140	41.8	recommended number
6	LP steam: rack header → HV-209 / XV-209 / TV-208 → E-203	6"-LS-302	6″	150 / A1S	0.70 / 175	31.5	yes

General notes

- Isometrics are not to scale: long runs are drawn shortened; lengths are centreline to centreline in metres and elevations (EL) are centreline elevations above grade, both true. North points up-right.
- Routes are those of the layout model (DC3-V1-PL-LAY-001 / layout.json); line numbers, sizes, classes and design conditions from the line list DC3-V1-PE-LIN-001; valve and instrument tags from the EPC P&ID DC3-V1-PID-001 rev D.
- Materials: hydrocarbon lines ASTM A333 Gr. 6 seamless (low-temperature carbon steel, impact tested to −46 °C), LP steam ASTM A106 Gr. B, Sch 40, ASME B16.5 weld-neck raised-face flanges, spiral-wound gaskets (assumed per the piping classes A1L / B1L / A1S).
- Supports generated by rule (assumed): rests at the suggested span for the size on horizontal runs (4″ 4.3 m, 6″ 5.2 m, 8″ 5.8 m), guides every 6 m on risers, rests / guides / line stops at the rack bents, a base support under each riser.
- Bill of materials: pipe = route length + 5 % cut and weld allowance; elbows counted at every change of direction; two flanges, gaskets and stud-bolt sets per flanged valve; one per equipment nozzle; instrument connections per tag.
- Hydrotest pressure 1.5 × design pressure (ASME B31.3 345.4.2, temperature correction ignored). Pump casing vents and drains, root valves of pressure instruments: typical, not shown (P&ID note 3).
- Symbols: bow-tie = gate (hand) valve with stem; with dome = control valve; with box "S" = SIS shut-off valve; half-filled = check valve; double bar = orifice flanges; square on a stem = pressure tap; circle on a stem = thermowell; diamond = sample connection.

DC3-V1-PI-ISO-001 sheet 1 · Feed: V-201 → P-201A/B → FV-201 station → E-201 → C-201 tray 18



▲ rest / base support ☐ guide ✕ anchor NTS: long runs drawn shortened; lengths (m, centreline to centreline) and EL (centreline) are true

Line	From → to	NPS	Class / spec	Material	Design P / T	MDMT	Test	Insulation	Length m
6"-P-201	V-201 bottom nozzle → P-201A/B suction	6″	ASME 150 / A1L	ASTM A333 Gr. 6 seamless (LT CS), Sch 40	1.35 MPa g / 65 °C	−46 °C	2.03 MPa g	AC	22.8
4"-P-202	P-201A/B discharge → E-201 channel (tube inlet)	4″	ASME 300 / B1L	ASTM A333 Gr. 6 seamless (LT CS), Sch 40	2.85 MPa g / 65 °C	−46 °C	4.28 MPa g	AC	46.2
6"-P-203	E-201 channel (tube outlet) → C-201 N1 (tray 18)	6″	ASME 300 / B1L	ASTM A333 Gr. 6 seamless (LT CS), Sch 40	2.85 MPa g / 175 °C	−46 °C	4.28 MPa g	H	26.7

Bill of materials · DC3-V1-PI-ISO-001 sheet 1

6"-P-201 · 6″ · ASTM A333 Gr. 6 seamless (LT CS) · ASME class 150 · Feed pump suction header

Item	Size	Qty	Unit	Note
Pipe, ASTM A333 Gr. 6 seamless (LT CS), Sch 40	6″	23.9	m	route length 22.7 m + 5 % (cut and weld allowance)
Elbow 90° LR, BW, Sch 40	6″	5	ea	
Tee, equal, BW	6″	1	ea	
Flange WN RF, class 150, bore to Sch 40	6″	7	ea	equipment nozzles 3 + 2 per flanged valve
Gasket spiral-wound, class 150	6″	7	ea	
Stud bolt set, class 150	6″	7	set	
Gate valve, class 150 (hand valve)	6″	2	ea	MV-201, MV-202
Support: Rest (shoe)	6″	2	ea	
Insulation AC (anti-condensation 40 mm (cold service at −1 °C, assumed))	6″	23.0	m	

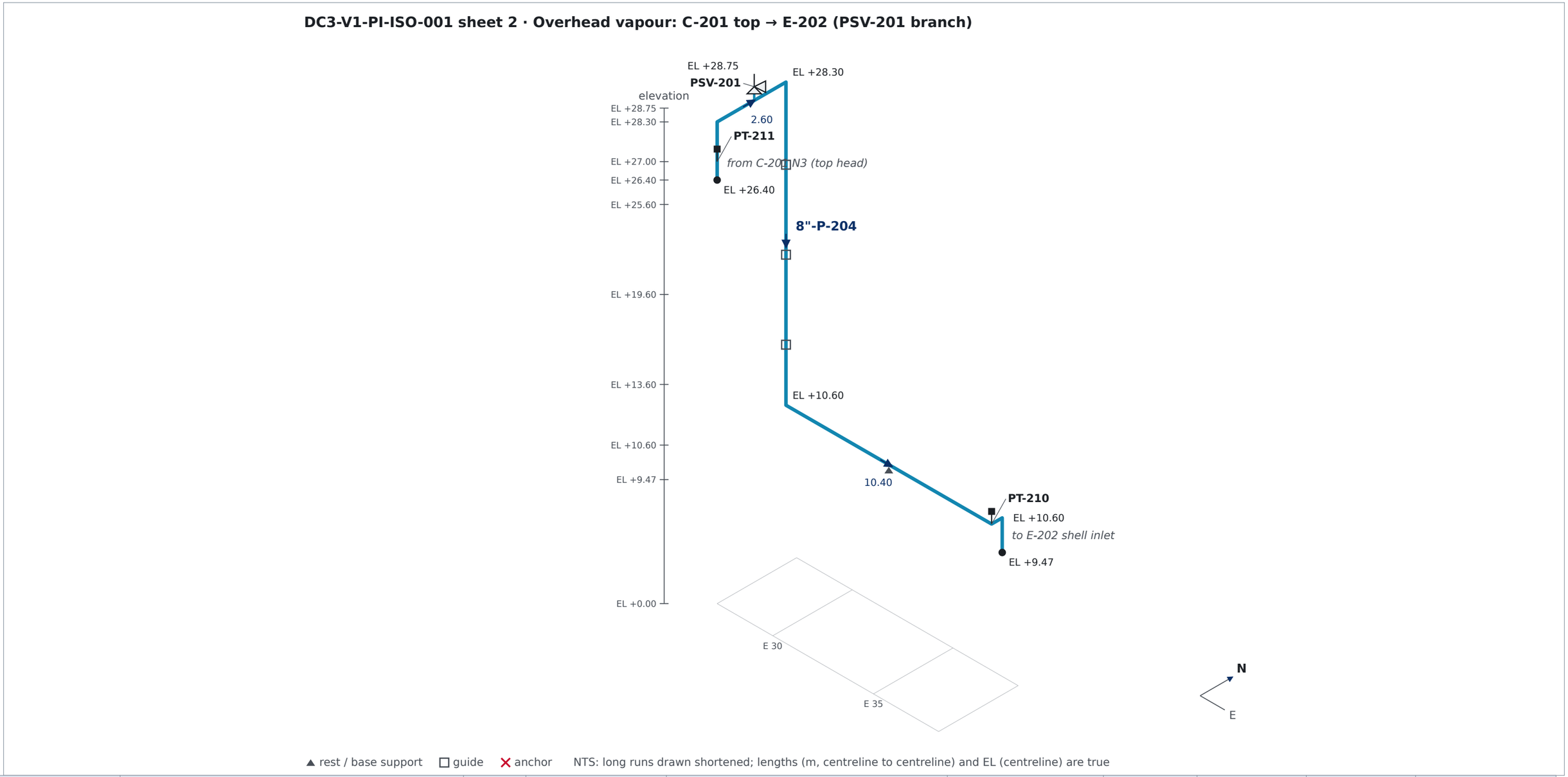
4"-P-202 · 4″ · ASTM A333 Gr. 6 seamless (LT CS) · ASME class 300 · Feed pump discharge to the preheater

Item	Size	Qty	Unit	Note
Pipe, ASTM A333 Gr. 6 seamless (LT CS), Sch 40	4″	48.6	m	route length 46.2 m + 5 % (cut and weld allowance)
Elbow 90° LR, BW, Sch 40	4″	15	ea	
Tee, equal, BW	4″	2	ea	
Flange WN RF, class 300, bore to Sch 40	4″	21	ea	equipment nozzles 3 + 2 per flanged valve
Orifice flange pair WN RF class 300 + plate	4″	1	set	
Gasket spiral-wound, class 300	4″	23	ea	
Stud bolt set, class 300	4″	23	set	
Gate valve, class 300 (hand valve)	4″	5	ea	MV-203, MV-205, MV-206, MV-204, MV-207
Control valve, class 300	4″	1	ea	FV-201
Shut-off valve, class 300, spring return	4″	1	ea	XV-201
Check valve, swing, class 300	4″	2	ea	NRV-201, NRV-202
Support: Rest (shoe)	4″	4	ea	
Insulation AC (anti-condensation 40 mm (cold service at −1 °C, assumed))	4″	47.0	m	

6"-P-203 · 6″ · ASTM A333 Gr. 6 seamless (LT CS) · ASME class 300 · Preheated feed (two-phase)

Item	Size	Qty	Unit	Note
Pipe, ASTM A333 Gr. 6 seamless (LT CS), Sch 40	6″	28.1	m	route length 26.7 m + 5 % (cut and weld allowance)
Elbow 90° LR, BW, Sch 40	6″	3	ea	
Flange WN RF, class 300, bore to Sch 40	6″	2	ea	equipment nozzles 2 + 2 per flanged valve
Gasket spiral-wound, class 300	6″	2	ea	
Stud bolt set, class 300	6″	2	set	
Thermowell 1½″ flanged	1½″	1	ea	TIC-202
Support: Rest (shoe)	6″	1	ea	
Support: Guide	6″	3	ea	
Support: Trunnion / base support	6″	1	ea	
Insulation H (heat conservation, mineral wool 50 mm (assumed))	6″	27.0	m	

DC3-V1-PI-ISO-001 sheet 2 · Overhead vapour: C-201 top → E-202 (PSV-201 branch)



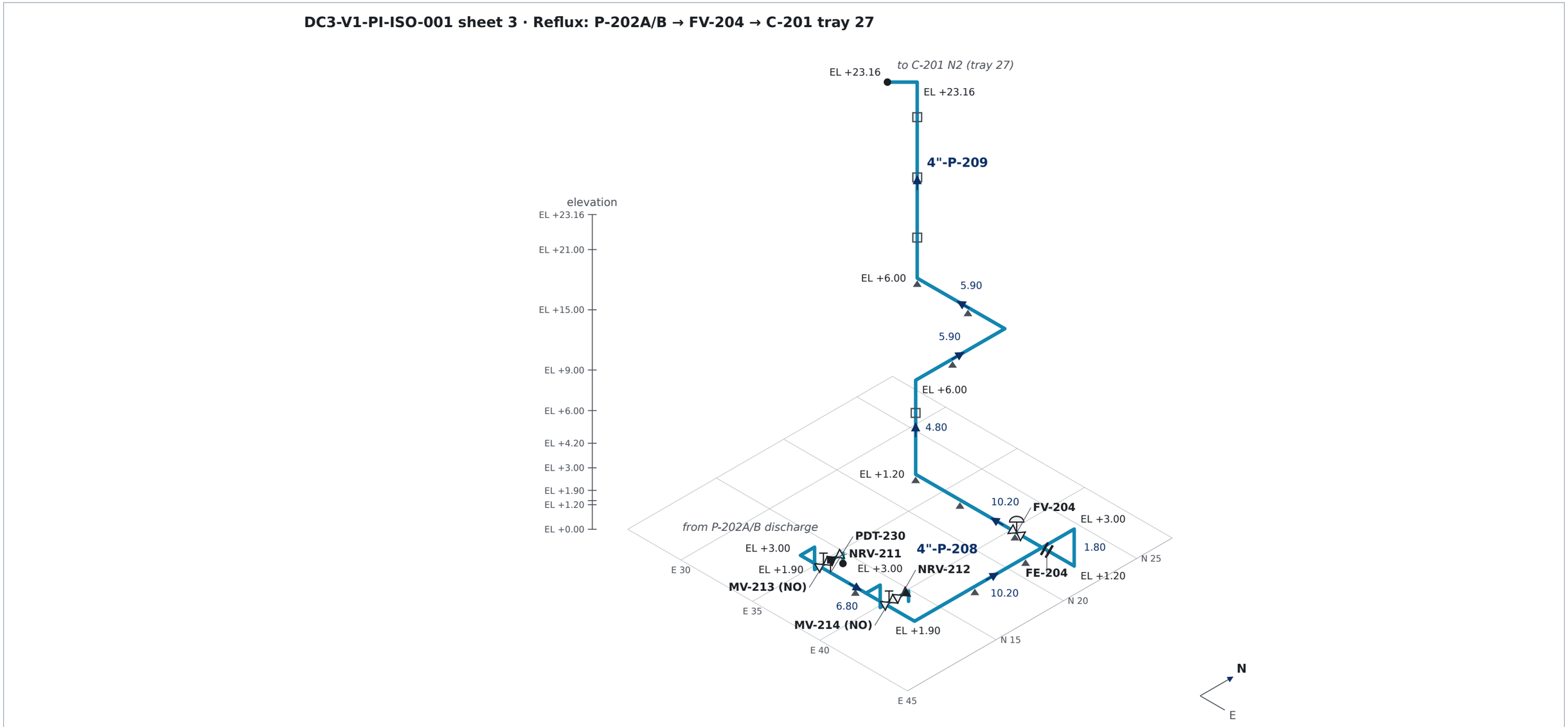
Line	From → to	NPS	Class / spec	Material	Design P / T	MDMT	Test	Insulation	Length m
8"-P-204	C-201 N3 (top head) → E-202 shell inlet	8″	ASME 300 / B1L	ASTM A333 Gr. 6 seamless (LT CS), Sch 40	1.90 MPa g / 75 °C	−46 °C	2.85 MPa g	—	34.6

Bill of materials · DC3-V1-PI-ISO-001 sheet 2

8"-P-204 · 8″ · ASTM A333 Gr. 6 seamless (LT CS) · ASME class 300 · Overhead vapour (PSV-201 branch)

Item	Size	Qty	Unit	Note
Pipe, ASTM A333 Gr. 6 seamless (LT CS), Sch 40	8″	36.4	m	route length 34.6 m + 5 % (cut and weld allowance)
Elbow 90° LR, BW, Sch 40	8″	5	ea	
Tee, reducing, BW	8″ × 4″	1	ea	
Flange WN RF, class 300, bore to Sch 40	8″	4	ea	equipment nozzles 3 + 2 per flanged valve
Gasket spiral-wound, class 300	8″	4	ea	
Stud bolt set, class 300	8″	4	set	
Relief valve (purchased by instruments)	8″	1	ea	PSV-201
Pressure tap: ¾″ weldolet + root valve	¾″	2	ea	PT-211, PT-210
Support: Rest (shoe)	8″	1	ea	
Support: Guide	8″	3	ea	

DC3-V1-PI-ISO-001 sheet 3 · Reflux: P-202A/B → FV-204 → C-201 tray 27



▲ rest / base support ☐ guide ✕ anchor NTS: long runs drawn shortened; lengths (m, centreline to centreline) and EL (centreline) are true

Line	From → to	NPS	Class / spec	Material	Design P / T	MDMT	Test	Insulation	Length m
4"-P-208	P-202A/B discharge → tee: reflux 4"-P-209 / product 3"-P-213	4″	ASME 300 / B1L	ASTM A333 Gr. 6 seamless (LT CS), Sch 40	2.60 MPa g / 70 °C	−46 °C	3.90 MPa g	—	26.8
4"-P-209	tee on 4"-P-208 → C-201 N2 (tray 27)	4″	ASME 300 / B1L	ASTM A333 Gr. 6 seamless (LT CS), Sch 40	2.60 MPa g / 70 °C	−46 °C	3.90 MPa g	—	45.1

Bill of materials · DC3-V1-PI-ISO-001 sheet 3

4"-P-208 · 4″ · ASTM A333 Gr. 6 seamless (LT CS) · ASME class 300 · Reflux / product pump discharge header

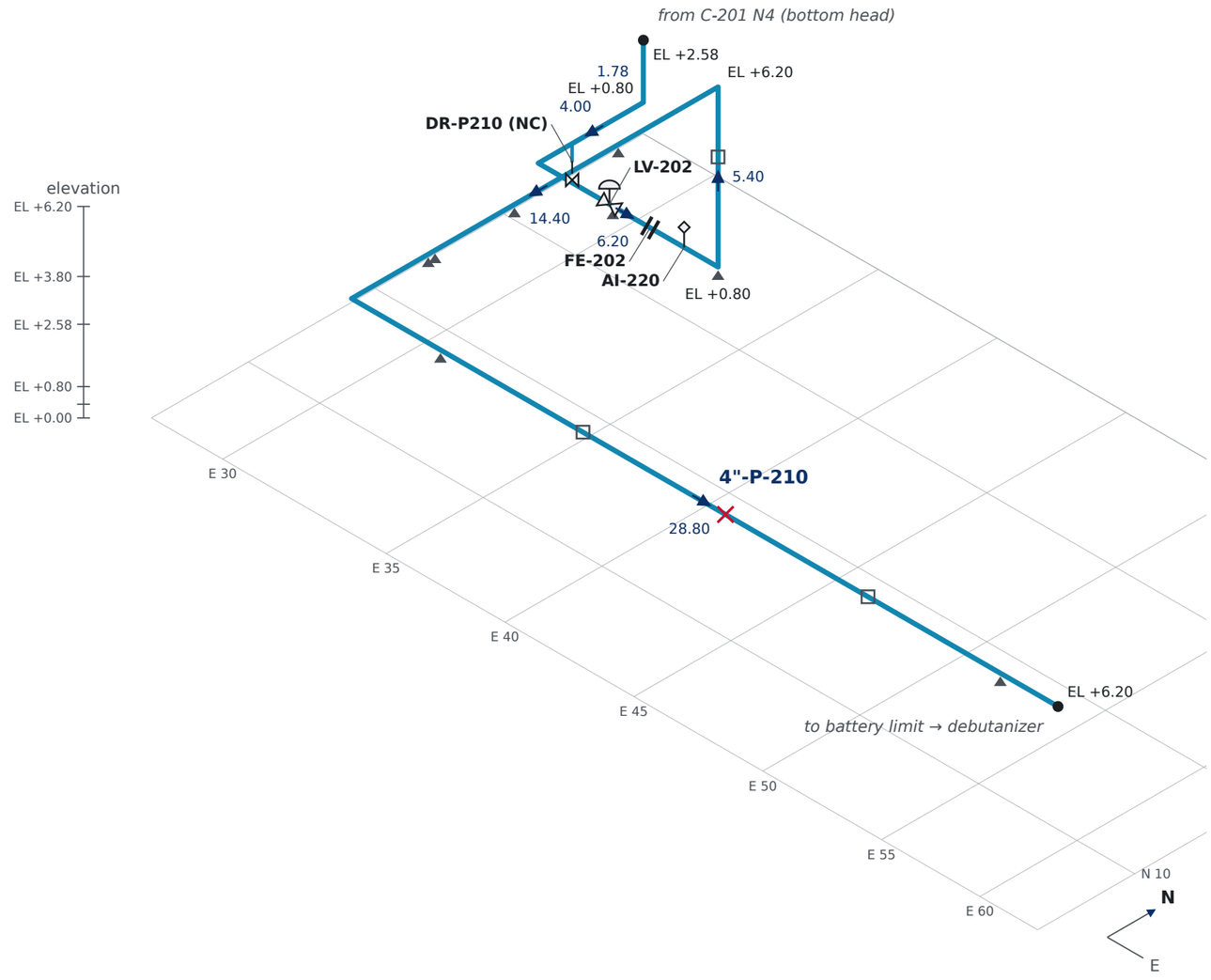
Item	Size	Qty	Unit	Note
Pipe, ASTM A333 Gr. 6 seamless (LT CS), Sch 40	4″	28.2	m	route length 26.8 m + 5 % (cut and weld allowance)
Elbow 90° LR, BW, Sch 40	4″	9	ea	
Tee, equal, BW	4″	1	ea	
Flange WN RF, class 300, bore to Sch 40	4″	10	ea	equipment nozzles 2 + 2 per flanged valve
Gasket spiral-wound, class 300	4″	10	ea	
Stud bolt set, class 300	4″	10	set	
Gate valve, class 300 (hand valve)	4″	2	ea	MV-213, MV-214
Check valve, swing, class 300	4″	2	ea	NRV-211, NRV-212
Pressure tap: ¾″ weldolet + root valve	¾″	1	ea	PDT-230
Support: Rest (shoe)	4″	3	ea	

4"-P-209 · 4″ · ASTM A333 Gr. 6 seamless (LT CS) · ASME class 300 · Reflux

Item	Size	Qty	Unit	Note
Pipe, ASTM A333 Gr. 6 seamless (LT CS), Sch 40	4″	47.5	m	route length 45.1 m + 5 % (cut and weld allowance)
Elbow 90° LR, BW, Sch 40	4″	5	ea	
Flange WN RF, class 300, bore to Sch 40	4″	3	ea	equipment nozzles 1 + 2 per flanged valve
Orifice flange pair WN RF class 300 + plate	4″	1	set	
Gasket spiral-wound, class 300	4″	5	ea	
Stud bolt set, class 300	4″	5	set	
Control valve, class 300	4″	1	ea	FV-204
Support: Rest (shoe)	4″	4	ea	
Support: Guide	4″	4	ea	
Support: Trunnion / base support	4″	2	ea	

DC3-V1-PI-ISO-001 sheet 4 · Bottoms: C-201 sump → LV-202 → rack → debutanizer

DC3-V1-PI-ISO-001 sheet 4 · Bottoms: C-201 sump → LV-202 → rack → debutanizer



▲ rest / base support ☐ guide ✕ anchor NTS: long runs drawn shortened; lengths (m, centreline to centreline) and EL (centreline) are true

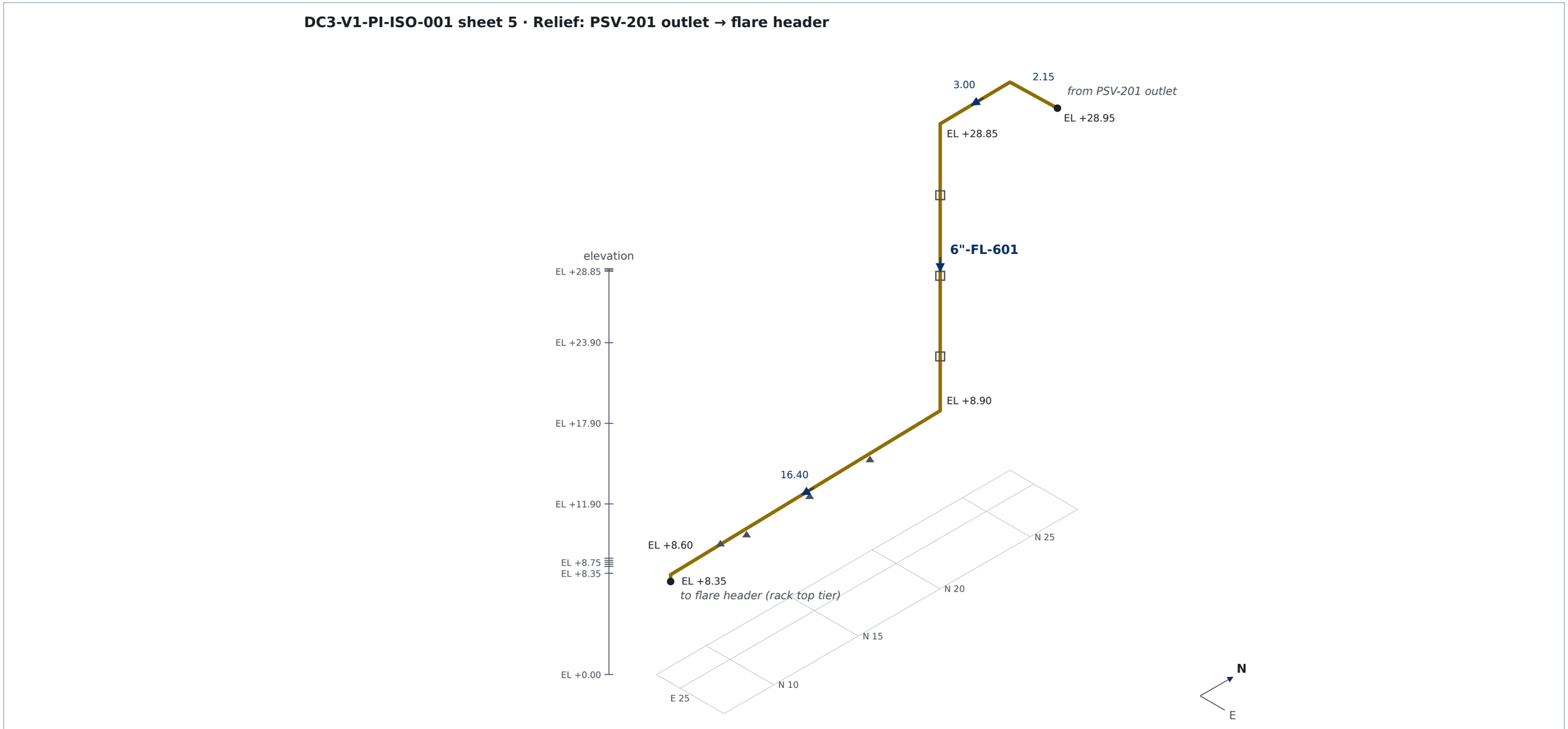
Line	From → to	NPS	Class / spec	Material	Design P / T	MDMT	Test	Insulation	Length m
4"-P-210	C-201 N4 (bottom head) → battery limit → debutanizer	4″	ASME 300 / B1L	ASTM A333 Gr. 6 seamless (LT CS), Sch 40	1.95 MPa g / 140 °C	−46 °C	2.93 MPa g	H	61.0

Bill of materials · DC3-V1-PI-ISO-001 sheet 4

4"-P-210 · 4″ · ASTM A333 Gr. 6 seamless (LT CS) · ASME class 300 · Bottoms on pressure to the debutanizer

Item	Size	Qty	Unit	Note
Pipe, ASTM A333 Gr. 6 seamless (LT CS), Sch 40	4″	64.1	m	route length 61.0 m + 5 % (cut and weld allowance)
Elbow 90° LR, BW, Sch 40	4″	5	ea	
Tee, reducing, BW	4″ × 2″	1	ea	
Flange WN RF, class 300, bore to Sch 40	4″	3	ea	equipment nozzles 1 + 2 per flanged valve
Orifice flange pair WN RF class 300 + plate	4″	1	set	
Gasket spiral-wound, class 300	4″	5	ea	
Stud bolt set, class 300	4″	5	set	
Control valve, class 300	4″	1	ea	LV-202
Sample connection: ¾″ weldolet + root valve	¾″	1	ea	AI-220
Drain: 2″ weldolet + gate valve + blind	2″	1	ea	DR-P210
Support: Rest (shoe)	4″	7	ea	
Support: Guide	4″	3	ea	
Support: Anchor (line stop)	4″	1	ea	
Support: Trunnion / base support	4″	1	ea	
Insulation H (heat conservation, mineral wool 50 mm (assumed))	4″	62.0	m	

DC3-V1-PI-ISO-001 sheet 5 · Relief: PSV-201 outlet → flare header



▲ rest / base support ☐ guide ✕ anchor NTS: long runs drawn shortened; lengths (m, centreline to centreline) and EL (centreline) are true

Line	From → to	NPS	Class / spec	Material	Design P / T	MDMT	Test	Insulation	Length m
6"-FL-601	PSV-201 outlet → flare header (rack top tier)	6″	ASME 150 / A1L	ASTM A333 Gr. 6 seamless (LT CS), Sch 40	0.50 MPa g / 140 °C	−46 °C	0.75 MPa g	—	41.8

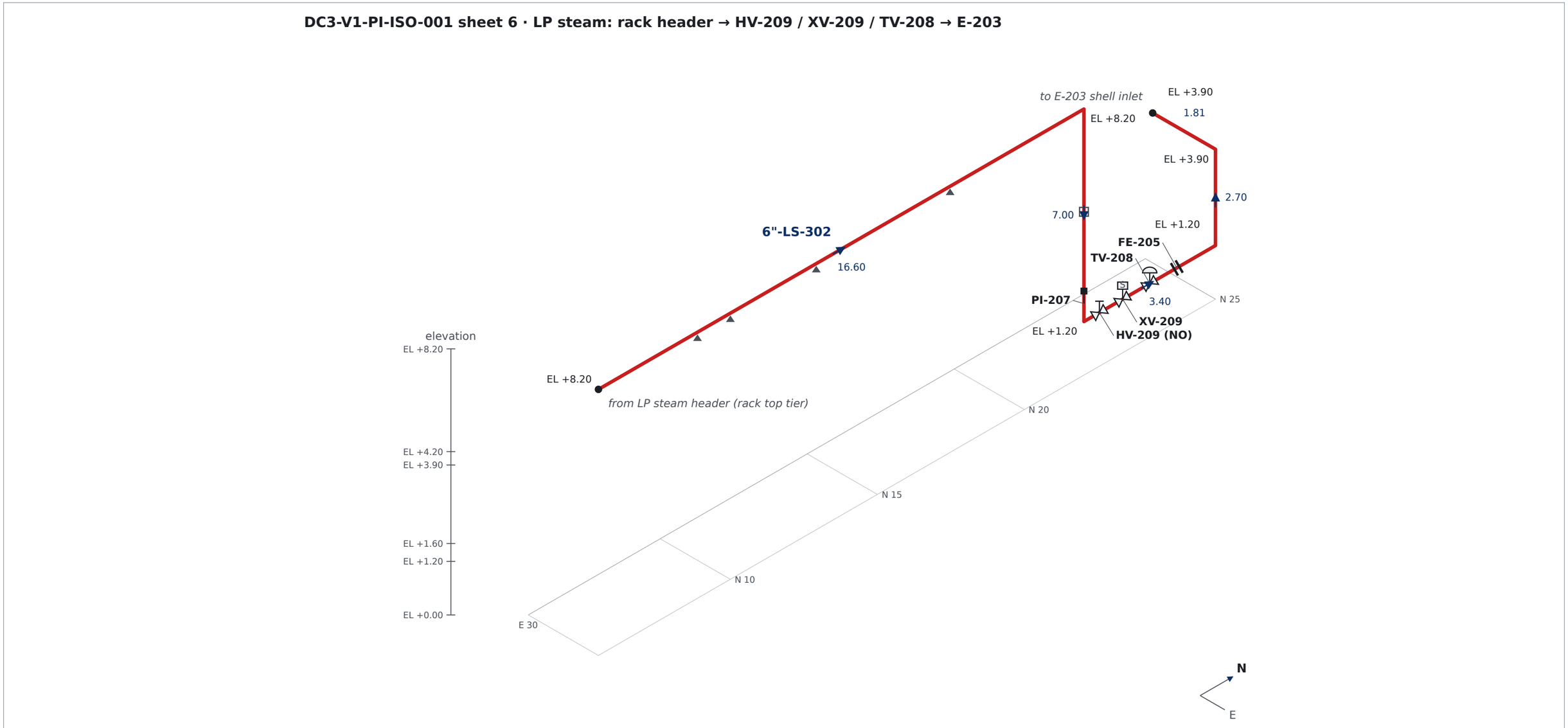
6″-FL-601 is not numbered on the P&ID (the outlet is drawn as "4″x6″" to flare): the number is recommended, query LA-01.

Bill of materials · DC3-V1-PI-ISO-001 sheet 5

6"-FL-601 · 6″ · ASTM A333 Gr. 6 seamless (LT CS) · ASME class 150 · PSV-201 discharge (tail pipe) to the closed flare header

Item	Size	Qty	Unit	Note
Pipe, ASTM A333 Gr. 6 seamless (LT CS), Sch 40	6″	43.9	m	route length 41.8 m + 5 % (cut and weld allowance)
Elbow 90° LR, BW, Sch 40	6″	4	ea	
Flange WN RF, class 150, bore to Sch 40	6″	1	ea	equipment nozzles 1 + 2 per flanged valve
Gasket spiral-wound, class 150	6″	1	ea	
Stud bolt set, class 150	6″	1	set	
Support: Rest (shoe)	6″	4	ea	
Support: Guide	6″	3	ea	

DC3-V1-PI-ISO-001 sheet 6 · LP steam: rack header → HV-209 / XV-209 / TV-208 → E-203



Line	From → to	NPS	Class / spec	Material	Design P / T	MDMT	Test	Insulation	Length m
6"-LS-302	LP steam header (rack top tier) → E-203 shell inlet	6″	ASME 150 / A1S	ASTM A106 Gr. B seamless, Sch 40	0.70 MPa g / 175 °C	0 °C	1.05 MPa g	H	31.5

Bill of materials · DC3-V1-PI-ISO-001 sheet 6

6"-LS-302 · 6″ · ASTM A106 Gr. B seamless · ASME class 150 · LP steam to the reboiler

Item	Size	Qty	Unit	Note
Pipe, ASTM A106 Gr. B seamless, Sch 40	6″	33.1	m	route length 31.5 m + 5 % (cut and weld allowance)
Elbow 90° LR, BW, Sch 40	6″	4	ea	
Flange WN RF, class 150, bore to Sch 40	6″	7	ea	equipment nozzles 1 + 2 per flanged valve
Orifice flange pair WN RF class 150 + plate	6″	1	set	
Gasket spiral-wound, class 150	6″	9	ea	
Stud bolt set, class 150	6″	9	set	
Gate valve, class 150 (hand valve)	6″	1	ea	HV-209
Control valve, class 150	6″	1	ea	TV-208
Shut-off valve, class 150, spring return	6″	1	ea	XV-209
Pressure tap: ¾″ weldolet + root valve	¾″	1	ea	PI-207
Support: Rest (shoe)	6″	4	ea	
Support: Guide	6″	1	ea	
Insulation H (heat conservation, mineral wool 50 mm (assumed))	6″	32.0	m	